

LINEAR ALGEBRA: VECTORS AND SPACES

DOT PRODUCT

- Determines the extent 2 vectors move in the same direction; 0 if $\perp$ and 1 if $\vec{a} \cdot \vec{b}$
- Also used to project one $\vec{v}$ onto another.

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

CROSS PRODUCT

Determines $\vec{c}$, orthogonal to $\vec{a}$ and $\vec{b}$

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \times \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix}$$

REDUCED ROW ECHELON

Reduce a space of vectors to a simpler form and determine # of solutions.

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & \dots & a \\ 0 & 1 & 0 & \dots & b \\ 0 & 0 & 1 & \dots & c \end{array} \right]$$

$$\begin{array}{l} |x_1 = a \\ |x_2 = b \\ |x_3 = c \end{array} \quad \begin{array}{l} \vdots \\ \vdots \\ \vdots \end{array} \quad \begin{array}{l} \text{Note: other than} \\ \text{1st element, rest} \\ \text{can be "anything"}. \end{array}$$

If bottom row all 0s:

→ $c = 0$: ∞ solutions

→ $c \neq 0$: no solution

Else: one solution

TERMS + PROPERTIES

Dot Product

$$\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cdot \cos(\theta)$$

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

LD/LI:

Linearly (in)dependent:

Can the vectors be expressed as a linear combination of the others?

Cross Product

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin(\theta)$$

Right-hand rule: 

$\vec{c}$ is cross-product direction

$$\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$$

Rank:

The # of LI vectors in the basis

NULL SPACE

All vectors $\vec{v}$ where, w/ Matrix **A**, $\mathbf{A} \vec{v} = \vec{0}$

1) augment **A** w/ $\vec{0}$:

$$\left[\begin{array}{cccc|c} a_1 & a_2 & \dots & a_n & 0 \\ b_1 & b_2 & \dots & b_n & 0 \\ c_1 & c_2 & \dots & c_n & 0 \end{array} \right]$$

2) First reduce row echelon form

3) Identify free vars (w/o leading 1s)

4) Solve for dependent vars in terms of free vars

5) create LI vectors by setting free vars to 1 and others to 0

6) Express null space as linear combination of these vectors

COLUMN SPACE

The subspace spanned by its column vectors; basis is a "smaller" vector space with only what it needs.

Includes all possible transformations of the vectors.

1) Identify cols in the RREF with pivots (leading 1s)

2) Corresponding cols in the original matrix form the basis

Note: vector $\vec{b}$ only belongs to the column space if

$$\mathbf{A} \vec{x} = \vec{b}$$